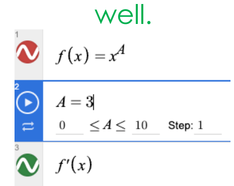
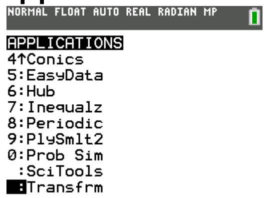
2.5 Applying the Power Rule
2.6 Derivative Rules: Constant, Sum, Difference, and Constant Multiple

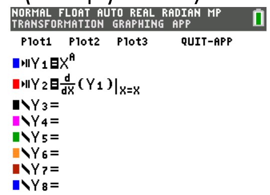
This activity can be done on Desmos as well.



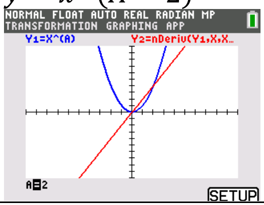
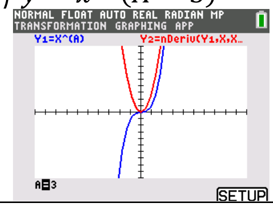
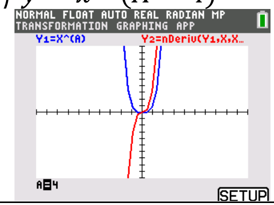
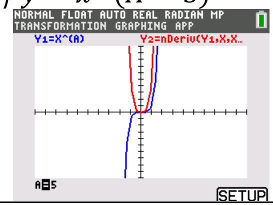
Ex 1: On your TI84 calculator, press **app** and scroll down (or up) until you find the Transfrm app and press enter. Press **y=** and type in:

$$Y_1 = x^A \qquad Y_2 = \frac{d}{dx}(Y_1)|_{x=x}$$





Press **graph**. Using the left and right arrows change the value of A to graph the derivatives of…

a) $y = x^2$ $(A = 2)$	b) $y = x^3$ $(A = 3)$	c) $y = x^4$ $(A = 4)$	d) $y = x^5$ $(A = 5)$
			

What relationship do you notice about the degrees of y and y'?

The degree of y' is one less than the degree of y.

Shortcut Rules

In the last lesson, you used the limit definition to find derivatives. In this and the next few, you will be introduced to several "shortcut rules" that allow you to find derivatives *without* the use of the limit definition.

1. Constant Rule: $\frac{d}{dx}[c] = 0$

Ex 2: Differentiate.

a) $y = 12345\pi$	b) $f(x) = \sqrt{9}$
$y' = 0$	$f'(x) = 0$

2. Power Rule: $\frac{d}{dx}[x^n] = nx^{n-1}$

Ex 3: Differentiate.

a) $f(x) = x^3$	b) $y = x^{20}$	c) $g(x) = \frac{1}{x}$
$f'(x) = 3x^2$	$y' = 20x^{19}$	$g(x) = x^{-1}$ $g'(x) = -x^{-2} = -\frac{1}{x^2}$

d) $h(x) = \sqrt{x}$	e) $f(x) = \frac{1}{\sqrt{x}}$
$h(x) = x^{\frac{1}{2}}$ $h'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}}$	$f(x) = x^{-\frac{1}{2}}$ $f'(x) = -\frac{1}{2}x^{-\frac{3}{2}} = -\frac{1}{2x^{\frac{3}{2}}}$

3. Constant Multiple Rule: $\frac{d}{dx}[cf(x)] = cf'(x)$

Ex 4: Differentiate.

a) $s(t) = 20t^7$

$s'(t) = 140t^6$

b) $y = \frac{4}{\sqrt[3]{x}}$

$y = 4x^{-\frac{1}{3}}$

$\frac{dy}{dx} = 4 \cdot -\frac{1}{3}x^{-\frac{4}{3}} = -\frac{4}{3x^{\frac{4}{3}}}$

c) $y = \pi$

$y' = 0$

d) $y = 10x$

$y' = 10$

*students find it easier to just read off the slope!

4. Sum and Difference Rules: $\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$

Ex 5: Differentiate.

a) $f(x) = 3x^4 - 2x + \pi$

$f'(x) = 12x^3 - 2$

b) $y = \pi^2 + \frac{1}{\pi} - \sqrt{\pi}$

$\frac{dy}{dx} = 0$

c) $g(x) = \frac{x}{5} - \frac{5}{x}$

$g(x) = \frac{1}{5}x - 5x^{-1}$

$g'(x) = \frac{1}{5} + 5x^{-2}$

$g'(x) = \frac{1}{5} + \frac{5}{x^2}$

d) $v(t) = (t+1)(3t-7)$

$v(t) = 3t^2 - 4t - 7$

$v'(t) = 6t - 4$

e) $y = (x-5)^2$

$y = x^2 - 10x + 25$

$y' = 2x - 10$

f) $y = \frac{3x^4 + 2x^2 - 5}{x^2}$

$y = 3x^2 + 2 - 5x^{-2}$

$\frac{dy}{dx} = 6x + 10x^{-3}$

$\frac{dy}{dx} = 6x + \frac{10}{x^3}$

g) For the gifted…

Let $\star \in \mathbb{Q}$

$$y = 5x^{\star} - \frac{2}{x^{\star - 1}}$$

$y = 5x^{\star} - 2x^{-\star + 1}$

$y' = 5\star x^{\star - 1} - 2(-\star + 1)x^{-\star}$

$y' = 5\star x^{\star - 1} - \frac{2(-\star + 1)}{x^{\star}}$

Ex 6: Write an equation of the tangent line to $f(x) = \frac{2}{\sqrt[5]{x}}$ at the point (1, 2).

$$f(x) = 2x^{-\frac{1}{5}}$$
$$f'(x) = -\frac{2}{5}x^{-\frac{6}{5}} = -\frac{2}{5x^{\frac{6}{5}}}$$
$$f'(1) = -\frac{2}{5}$$
$$y - 2 = -\frac{2}{5}(x - 1)$$

Ex 7: Determine the points, if any, at which $g(x) = 2x^3 + 3x^2 - 12x + 5$ has a horizontal tangent line.

$$g'(x) = 6x^2 + 6x - 12$$
$$g'(x) = 6(x^2 + x - 2)$$
$$g'(x) = 6(x + 2)(x - 1) = 0$$
$$x = -2, 1$$
$$g(-2) = 25 \qquad g(1) = -2$$

Ex 8:

Let f be the function given by $f(x) = x^3 - 6x^2 + 8x - 2$. What is the instantaneous rate of change of f at $x = 3$?

(A) -5 (B) $-\frac{15}{4}$ (C) -1 (D) 6 (E) 17

$$f'(x) = 3x^2 - 12x + 8$$
$$f'(3) = -1$$

Ex 9:

$$\lim_{x \to e} \frac{(x^{20} - 3x) - (e^{20} - 3e)}{x - e} \text{ is}$$

(A) 0 (B) $20e^{19} - 3$ (C) $e^{20} - 3e$ (D) nonexistent

$$\lim_{x \to e} \frac{(x^{20} - 3x) - (e^{20} - 3e)}{x - e} = \frac{d}{dx}(x^{20} - 3x)\Big|_{x=e}$$
$$\frac{d}{dx}(x^{20} - 3x) = 20x^{19} - 3$$
$$\frac{d}{dx}(x^{20} - 3x)\Big|_{x=e} = 20e^{19} - 3$$

Ex 10: Let f be the piecewise-linear function defined. Which of the following statements are true?

$$f(x) = \begin{cases} 2x - 2 & for\ \ x < 3 \\ 2x - 4 & for\ \ x \geq 3 \end{cases}$$

I. $\lim_{h \to 0^-} \frac{f(3+h) - f(3)}{h} = 2$

II. $\lim_{h \to 0^+} \frac{f(3+h) - f(3)}{h} = 2$

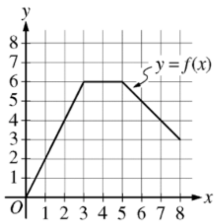
III. $f'(3) = 2$

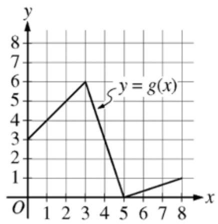
(A) None
(B) II only
(C) I and II only
(D) I, II and III

$$\lim_{x \to 3^-} f(x) \neq \lim_{x \to 3^+} f(x)$$

$f(x)$ is NOT continuous at $x = 3$.

Ex 11: The graphs of the piecewise linear functions f and g are shown.





a) The function h is defined by $h(x) = f(x) + 3g(x)$, find $h'(2)$ or state why it doesn't exist.

$$h'(x) = f'(x) + 3g'(x)$$
$$h'(2) = f'(2) + 3g'(2)$$
$$h'(2) = 2 + 3(1)$$

b) The function p is defined by $p(x) = 6x^2 - 5\,f(x)$, find $p'(3)$ or state why it doesn't exist.

$$p'(x) = 12x - 5f'(x)$$

$f'(3)$ DNE due to a sharp turn, $\therefore$ $p'(3)$ DNE

Ex 12: The functions $f(x)$ and $g(x)$ are differentiable. The table gives values of the functions and their first derivatives at selected values of x.

x	0	2	4	7
$f(x)$	10	7	4	5
$f'(x)$	$\frac{3}{2}$	-8	3	6
$g(x)$	1	2	-3	0
$g'(x)$	5	4	2	8

a) Write an equation of the tangent line to $m(x) = 2f(x) + 3$, at $x = 7$.

$$m(7) = 2f(7) + 3 = 2(5) + 3 = 13$$
$$m'(x) = 2f'(x)$$
$$m'(7) = 2f'(7) = 2(6) = 12$$
$$y - 13 = 12(x - 7)$$

b) The function n is defined by $n(x) = f(x) + 2g(x)$. Determine the x –values, if any, at which $n(x)$ has a horizontal tangent line.

$$n(x) = f(x) + 2g(x)$$
$$n'(x) = f'(x) + 2g'(x) = 0$$
$$x = 2$$

Ex 13: Find an equation of the line that is tangent to $f(x) = 5x^2 - 3$ and parallel to $5x - y = 4$.

Parallel = same slope

The slope of $5x - y = 4$ is 5.

The slope of $f(x)$ is $f'(x) = 10x$

$$10x = 5$$
$$x = \frac{1}{2}$$
$$f\left(\frac{1}{2}\right) = -\frac{7}{4}$$
$$y + \frac{7}{4} = 5\left(x - \frac{1}{2}\right)$$

Ex 14: Find k such that the line $y = -2x + 3$ is tangent to $f(x) = kx^2$.

Functions and tangent lines share a point and slope at the point of tangency.

Same point	Same slope
$-2x + 3 = kx^2$	$-2 = 2kx$

$$-2 = 2kx \implies x = -\frac{1}{k}$$
$$-2\left(-\frac{1}{k}\right) + 3 = k\left(-\frac{1}{k}\right)^2$$
$$\frac{2}{k} + 3 = \frac{1}{k}$$
$$\frac{1}{k} = -3$$
$$k = -\frac{1}{3}$$